

Assignment 2:

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Microsoft Windows [Version 10.0.26200.8246]  
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```
C:\Users\neera>ping yahoo.com
```

```
Pinging yahoo.com [2001:4998:24:120d::1:0] with 32 bytes of data:
```

```
Reply from 2001:4998:24:120d::1:0: time=284ms
```

```
Reply from 2001:4998:24:120d::1:0: time=282ms
```

```
Reply from 2001:4998:24:120d::1:0: time=283ms
```

```
Reply from 2001:4998:24:120d::1:0: time=284ms
```

```
Ping statistics for 2001:4998:24:120d::1:0:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
    Approximate round trip times in milli-seconds:
```

```
        Minimum = 282ms, Maximum = 284ms, Average = 283ms
```

Capture and TCP analysis:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	140.82.112.26	172.16.0.2	TCP	40	443 → 49670 [ACK] Seq=1 Ack=1 Win=18 Len=0
2	0.000085700	172.16.0.2	140.82.112.26	TCP	40	[TCP ACKed unseen segment] 49670 → 443 [ACK] Seq=1
3	0.905614200	140.82.112.26	172.16.0.2	TLSv1.2	65	[TCP Previous segment not captured] , Application
4	0.906452700	172.16.0.2	140.82.112.26	TLSv1.2	69	Application Data
5	0.941414900	140.82.112.26	172.16.0.2	TCP	40	443 → 49670 [ACK] Seq=27 Ack=30 Win=18 Len=0
8	1.793740000	20.207.73.82	172.16.0.2	TCP	40	443 → 57176 [ACK] Seq=1 Ack=1 Win=17 Len=0
9	1.793857300	172.16.0.2	20.207.73.82	TCP	40	[TCP ACKed unseen segment] 57176 → 443 [ACK] Seq=1
12	2.258359800	172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21795/9045, tt
13	2.292018400	162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21795/9045, tt
14	3.374630100	20.207.73.82	172.16.0.2	TLSv1.2	79	[TCP Previous segment not captured] , Application
15	3.375739300	172.16.0.2	20.207.73.82	TLSv1.2	64	Application Data
16	3.375828500	172.16.0.2	20.207.73.82	TCP	40	57176 → 443 [FIN, ACK] Seq=25 Ack=41 Win=251 Len=0
17	3.409073000	20.207.73.82	172.16.0.2	TCP	40	443 → 57176 [ACK] Seq=41 Ack=25 Win=17 Len=0
18	3.409196000	20.207.73.82	172.16.0.2	TCP	40	443 → 57176 [FIN, ACK] Seq=41 Ack=26 Win=17 Len=0
19	3.409563700	172.16.0.2	20.207.73.82	TCP	40	57176 → 443 [ACK] Seq=26 Ack=42 Win=251 Len=0
28	4.867769200	98.70.192.224	172.16.0.2	TCP	40	443 → 64770 [ACK] Seq=1 Ack=1 Win=18 Len=0
29	4.867841800	172.16.0.2	98.70.192.224	TCP	40	[TCP ACKed unseen segment] 64770 → 443 [ACK] Seq=1

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Capture and DNS analysis:

No.	Time	Source	Destination	Protocol	Length	Info
43	10.410230700	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x3869 A www.bing.com OPT
44	10.411003800	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x7cbf AAAA www.bing.com OPT
45	10.411299900	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x2caa A www.bing.com OPT
46	10.438488600	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x6620 AAAA www.bing.com OPT
47	10.438488600	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x3e61 A www.bing.com OPT
48	10.438505000	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x67dc A www.bing.com OPT
49	10.446498100	162.159.36.1	172.16.0.2	DNS	330	Standard query response 0x1c88 AAAA www.bing.com C
50	10.446691200	162.159.36.1	172.16.0.2	DNS	330	Standard query response 0x7cbf AAAA www.bing.com C
51	10.475843300	162.159.36.1	172.16.0.2	DNS	330	Standard query response 0x6620 AAAA www.bing.com C
52	10.510761300	162.159.36.1	172.16.0.2	DNS	334	Standard query response 0x2caa A www.bing.com CNAM
53	10.510950400	162.159.36.1	172.16.0.2	DNS	334	Standard query response 0x3e61 A www.bing.com CNAM
54	10.511001000	162.159.36.1	172.16.0.2	DNS	334	Standard query response 0x67dc A www.bing.com CNAM
55	10.511036000	162.159.36.1	172.16.0.2	DNS	334	Standard query response 0x3869 A www.bing.com CNAM
293	21.327481200	172.16.0.2	162.159.36.1	DNS	76	Standard query 0x533f A telemetry.canva.com OPT
294	21.327792900	172.16.0.2	162.159.36.1	DNS	76	Standard query 0x9cee A telemetry.canva.com OPT
295	21.327813000	172.16.0.2	162.159.36.1	DNS	76	Standard query 0x715f AAAA telemetry.canva.com OPT

> Frame 42: Packet, 69 bytes on wire (552 bits), 69 bytes captured | 0000 45 00 00 45 89 48 00 00 80 11 3e ad ac 10 00 02 E..E.H..

Capture and UDP analysis:

No.	Time	Source	Destination	Protocol	Length	Info
37	7.119172100	2606:4700:110:8863:...	2404:6800:4009:803:...	UDP	77	54365 → 443 Len=29
42	10.409589100	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x3869 A www.bing.com OPT
43	10.410230700	172.16.0.2	162.159.36.1	DNS	69	Standard query 0x1c88 AAAA www.bing.com OPT

Capture and ICMP analysis:

The figure displays a Wireshark packet capture of ICMP Echo (ping) traffic. The packet list shows 39 packets, with packet 39 selected. The packet details pane shows the structure of the ICMP Echo (ping) request, including the type, code, and sequence number.

No.	icmp	Source	Destination	Protocol	Length	Info
13	icmpv6	172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21795/9045, tt
18		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21795/9045, tt
37		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21796/9301, tt
39		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21796/9301, tt
199		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21797/9557, tt
200		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21797/9557, tt
258		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21798/9813, tt
259		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21798/9813, tt
310		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21799/10069, tt
311		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21799/10069, tt
328		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21800/10325, tt
329		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21800/10325, tt
352		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21801/10581, tt
353		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21801/10581, tt
388		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21802/10837, tt
389		162.159.198.2	172.16.0.2	ICMP	60	Echo (ping) reply id=0x0001, seq=21802/10837, tt
402		172.16.0.2	162.159.198.2	ICMP	60	Echo (ping) request id=0x0001, seq=21803/11093, tt

Frame 39: Packet, 60 bytes on wire (480 bits), 60 bytes captured
 Raw packet data
 Internet Protocol Version 4, Src: 162.159.198.2, Dst: 172.16.0.2
 Internet Control Message Protocol

The image shows a Wireshark packet capture window. The top pane displays the details of Frame 18, which is a DNS query packet. The packet is 69 bytes on wire and 69 bytes captured. It is an Internet Protocol Version 4 packet from 172.16.0.2 to 162.159.36.1, using User Datagram Protocol (UDP) port 19894 to port 53. The domain name system query is for 'www.bing.com'.

The bottom pane shows the raw packet data in hexadecimal and ASCII. The hexadecimal data is: 45 00 00 45 8d 04 00 00 80 11 3a f1 ac 10 00 02 a2 9f 24 01 4d b6 00 35 00 31 48 00 dc b2 01 00 00 01 00 00 00 00 00 01 03 77 77 77 04 62 69 6e 67 03 63 6f 6d 00 00 01 00 01 00 00 29 04 d0 00 00 00 00 00 00. The ASCII data is: E..E.... :.:... ..\$.M..5..1H.... :..... www bin g.com.... :....

The status bar at the bottom indicates: No.: 18 - Time: 4.679286000 - Source: 172.16.0.2 - Destination: 162.159.36.1 - Protocol: DNS - Length: 69 - Info: Standard query 0x4cb2 A www.bing.com OPT. The 'Show packet bytes' checkbox is checked, and the layout is set to 'Vertical (Stacked)'.

Offset	Hex	ASCII
0000	45 00 00 45 8d 04 00 00	E..E.... :.:...
0010	a2 9f 24 01 4d b6 00 35	..\$.M..5..1H....
0020	00 01 00 00 00 00 00 01	:..... www bin
0030	03 77 77 77 04 62 69 6e	g.com.... :....
0040	67 03 63 6f 6d 00 00 01	
0050	00 01 00 00 29 04 d0 00	
0060	00 00 00 00 00 00 00 00	